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Hardware-Software Co-design, Numerical Optimization

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PAPER PUBLICATIONS

• **Causal-History Test-Time Scaling for Failure Recovery in Autoregressive World-Action Models**

*Lin Li, Long Chen, **Kwunhang Wong**, Jiaming Lei, Song Jin, Shucheng Du, Chuhan Zhang, Songchen Ma, Weihao Zhang, Jun Xiao, Kwang-Ting (Tim) Cheng*

– Submitted to International Conference on Robotics and Automation (ICRA'27)

• **AutoIsing: An Automated Compiler for Programmable Ising Machine**

*Yi Deng, Zihao Xuan, Ning Lin, **Kwunhang Wong**, Han Wang, and Zhongrui Wang*

– In Asia and South Pacific Design Automation Conference (ASP-DAC'27), Tokyo

– Accepted date: 04 Sep 2026 (Acceptance rate: 276/935=29.5%)

• **Weight-Stationary Fault Recovery via Neuron Tuning in RRAM-Based CIM Neuromorphic Accelerators**

*Yangu He, Xinyuan Zhang, Yuxi Chen, **Kwunhang Wong**, Zijian Ye, Chenchen Ding, Zhengwu Liu, Ning Lin, Chaoliang Tan, Xiaojuan Qi, Zhongrui Wang, and Han Wang*

– In Asia and South Pacific Design Automation Conference (ASP-DAC'27), Tokyo

– Accepted date: 04 Sep 2026 (Acceptance rate: 276/935=29.5%)

• **NeuronUnlearn: Machine Unlearning for RRAM-Based Computing-in-Memory Accelerators**

*Yangu He, Xinyuan Zhang, Jichang Yang, Yue Zhang, **Kwunhang Wong**, Bo Wang, Ning Lin, Xiaojuan Qi, Zhongrui Wang, and Han Wang*

– In International Conference on Computer Design (ICCD'26), Hong Kong

– Accepted date: 20 Aug 2026 (Acceptance rate: 26%)

• **RRAM-DP: Device-Calibrated Differential Privacy for In-Memory Edge Learning**

***Kwunhang Wong**, Jichang Yang, Karl M.H. Lai, Hegan Chen, Songqi Wang, Wei Xuan, Ning Lin, Han Wang, Xiaojuan Qi, and Zhongrui Wang*

– In International Conference on Computer-Aided Design (ICCAD'26), San Jose

– Accepted date: 11 Jul 2026 (<https://arxiv.org/pdf/2607.18169>) (Acceptance rate: 371/1465=25.3%)

• **Potentials and Limitations on Different Busbar Protections in Industrial Applications**

*Karl MH Lai, Yunhe Hou, and **Kwunhang Wong***

– In International Council on Electrical Engineering Conference (ICEE'26), Seoul

– Accepted date: 05 Jul 2026 (<https://arxiv.org/pdf/2607.28453>)

• **Comparative Analysis on Inertia Estimation Algorithms in Providing Proper Frequency Response**

*Karl MH Lai, Yunhe Hou, and **Kwunhang Wong***

– In International Council on Electrical Engineering Conference (ICEE'26), Seoul

– Accepted date: 05 Jul 2026 (<https://arxiv.org/pdf/2609.16254>)

• **Towards Secure and Efficient DNN Accelerators via Hardware-Software Co-Design**

*Wei Xuan, Zihao Xuan, Rongliang Fu, Ning Lin, **Kwunhang Wong**, Zikang Yuan, Lang Feng, Zhongrui Wang, Tsung-Yi Ho, Yuzhong Jiao, and Luhong Liang*

– Journal: IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)

– Accepted date: 02 Jun 2026 (<https://doi.org/10.1109/tcad.2026.3699292>)

• **AgeBalance: Low-Cost Lifetime Extension for SRAM-based PIM Accelerators**

*Ning Lin, Shaocong Wang, Yaping Zhao, Yangu He, Songqi Wang, **Kwunhang Wong**, Rongliang Fu, Wenxing Li, Tsung-Yi Ho, Dashan Shang, Xiaojuan Qi, Xiaoming Chen, and Zhongrui Wang*

– Journal: IEEE Transactions on Computers (TC)

– Accepted date: 20 May 2026 (<https://doi.org/10.1109/TC.2026.3695104>)

- **A Fully Analog Continuous-Time CIM Neural ODE Solver for Fluid Dynamics Generation**
*Songqi Wang, Meng Xu, Jichang Yang, Zhexu Chen, Hegan Chen, Sishuo Liu, Xinyuan Zhang, **Kwunhang Wong**, Ning Lin, Yi Li, Zhongrui Wang, and Han Wang*
 - In Design Automation Conference (DAC’26), Long Beach
 - Accepted date: 09 Mar 2026 (Acceptance rate: 22%)
- **Resistive Memory based Efficient Machine Unlearning and Continual Learning**
*N. Lin, J. Yang, Y. He, Z. Ye, **Kwunhang Wong**, X. Zhang, S. Wang, Y. Li, K. Xu, L. Zhang, X. Chen, D. Shang, H. Wang, X. Qi, and Z. Wang*
 - Journal: Nature Communication (in final revision process)
 - Accepted date: NULL (<https://arxiv.org/pdf/2601.10037v1>)
- **When Pipelined In-Memory Accelerators Meet Spiking Direct Feedback Alignment: A Co-Design**
*Haoxiong Ren, Yangu He, **Kwunhang Wong**, Rui Bao, Ning Lin, Zhongrui Wang, and Dashan Shang*
 - In International Conference on Computer-Aided Design (ICCAD’25), Munich
 - Accepted date: 30 Jun 2025 (<https://arxiv.org/abs/2507.15603>) (Acceptance rate: 266/1,078=24.7%)
- **Re4PUF: A Reliable, Reconfigurable ReRAM-based PUF Resilient to DNN and Side Channel Attacks**
*N. Lin, Y. Li, Y. He, S. Wang, H. Chen, **Kwunhang Wong**, C. Li, J. Yang, Y. Yu, M. Xu, Y. Han, R. Chen, X. Chen, X. Xu, J. Yang, D. Shang, and Z. Wang*
 - In Design Automation Conference (DAC’25), San Francisco
 - Accepted date: 26 Feb 2025 (<https://doi.org/10.1109/DAC63849.2025.11133290>) (Acceptance rate: 420/1,862=22.6%)
- **Resistive memory-based zero-shot liquid state machine for multimodal event data learning**
*N. Lin, S. Wang, Y. Li, B. Wang, S. Shi, Y. He, W. Zhang, Y. Yu, Y. Zhang, X. Zhang, **Kwunhang Wong**, S. Wang, X. Qi, X. Chen, H. Jiang, X. Zhang, P. Lin, X. Xu, Q. Liu, Z. Wang, D. Shang, and M. Liu*
 - Journal: Nature Computational Science
 - Accepted date: 26 Nov 2024 (<https://doi.org/10.1038/s43588-024-00751-z>)
- **SNNGX: Securing SNNs with Genetic XOR Encryption on RRAM-based Neuromorphic Accelerator**
***Kwunhang Wong**, Songqi Wang, Wei Huang, Xinyuan Zhang, Yangu He, Karl M.H. Lai, Yuzhong Jiao, Ning Lin, Xiaojuan Qi, Xiaoming Chen, and Zhongrui Wang*
 - In International Conference on Computer-Aided Design (ICCAD’24), New Jersey
 - Accepted date: 30 Jun 2024 (<https://doi.org/10.1145/3676536.3676727>) (Acceptance rate: 195/1,109=17.6%)
- **Older and Wise: The Marriage of Device Aging and Intellectual Property Protection of DNNs**
*Ning Lin, Shaocong Wang, Yue Zhang, Yangu He, **Kwunhang Wong**, Arindam Basu, Dashan Shang, Xiaoming Chen, and Zhongrui Wang*
 - In Design Automation Conference (DAC’24), San Francisco
 - Accepted date: 26 Feb 2024 (<https://doi.org/10.1145/3649329.3656223>) (Acceptance rate: 263/1156=22.7%)
- **In-memory and In-Sensor Reservoir Computing with Memristive Devices**
*Ning Lin, Jia Chen, Ruoyu Zhao, Yangu He, **Kwunhang Wong**, Qinru Qiu, Zhongrui Wang, and J. Joshua Yang*
 - Journal: APL Machine Learning (Selected as Journal Cover)
 - Accepted date: 10 Jan 2024 (<https://doi.org/10.1063/5.0174863>)

EDUCATION

- **PhD in Electrical and Computer Engineering (ECE), HKU** **2023-now**
Co-Supervised by Prof.Xiaojuan Qi and Prof.Zhongrui Wang CGPA: 4.0
Co-authored 17+ CS/CE papers publication: (DAC, ICCAD, Nature Computational Science)
- **BEng in Electrical Engineering (EE), HKU** **2018-2023**
Course taken: Advanced Robotics, Convex Optimization, Digital Electronics, Power Electronics
Thesis: Evolutionary Adversarial Attacks on the Inherent Robustness of Spiking Neural Networks
CLP Scholarship in Electrical Engineering, Dean’s honours list, Reaching Out Award 2019-2020
UCL, Faculty of Brain Sciences Summer Exchange (grade: 1st) 2018-2019

WORK EXPERIENCE

- **AI Chip Center for Emerging Smart Systems (ACCESS), HKUST** 02/2024-now
PhD Student Researcher Hong Kong
 - RP1.2: Computing-in-Memory enabled on-device Large Language Model Personalization
 - RP4.2: Design, Test, and Automation for Secure Machine Learning Accelerator
- **Department of Electrical and Computer Engineering, HKU** 06/2021-09/2021
Research Assistant Hong Kong
 - Conservation Voltage Reduction (CVR) application in power industry
 - Mathematical modelling of CVR Calculation supervised by **Prof. Jin Zhong**
- **CLP Power Hong Kong Limited** 06/2020-06/2021
Gap-year Internship Hong Kong
 - Designed 132/11kV substation network voltage-step-down load experiment
 - Proposed Volt-Var Optimization (VVO) study to mitigate reverse power flow dynamics in high-penetration distributed photovoltaic CLP Power grid

RELATED SIDE-PROJECTS

- **Real-Time Diffusion Transformer (DiT) Acceleration for Embodied AI** 2026
 - Accelerated diffusion-based robotic manipulation policies, achieving 2.09 ms low-latency inference.
 - Achievement: 8.75x speedup over RTX 4070 GPU with a 91.7% real-world task success rate.
 - Methods: 4-bit weight quantization for DiT architectures, iterative denoising control optimization, FPGA data movement scheduling, RRAM/DDR memory bandwidth mapping.
- **GPT-Astra based In-context Policy Learning on One-shot Retrieval** 2026
 - Designed episodic memory and LLM harness pipeline for GPT-Astra to execute one-shot policy adaptation
 - Achievement: success in 3 different manipulations, including alphabet block stacking, pick-and-place, floristry
 - Methods: integration with CLIP embeddings and RoboHarness on Codex closed-loop code-as-policy evaluations

TECHNICAL SKILLS AND INTERESTS

Languages: Cantonese, Mandarin, English, Japanese (N3)

Developer Tools: Codex, Obsidian, LaTeX, PyTorch, CUDA, Matlab, ROS, Python/C++/Java etc., Verilog

Technical Knowledge: AI Agent Workflow, LLM, Optimization, RAG, Reinforcement Learning, Differential Privacy

Areas of Interest: LLM Agents, Steerable Diffusion, On-Policy Distillation, Self-Evolving Algorithms/Systems

PROFESSIONAL SERVICE

Peer Reviewer (Machine Learning): AAAI (2024), ICLR (2024), ICML (2024), npj Artificial Intelligence (2025)

Peer Reviewer (Hardware & EDA): ICCAD (2024, 2026), ICCD (2026)

POSITIONS OF RESPONSIBILITY

- **External Vice-Chairperson, HKU Golf Club (Founding Committee)** 2023-2025
- **Python Summer Course Coordinator, SKH Bishop Baker Secondary School** 2023-now
- **Winter 2022 Startup Entrepreneur, Hong Kong innoX Academy (HKSTP x HKUST)** 2021
- **Team Captain, Ricci Hall Sports Team (Aquatics, Basketball, Softball)** 2019, 2023, 2024

ACHIEVEMENTS

- **IEEE Blockchain HK Web3.0 Developer Hackathon 2022 (Most Innovative Award)** 08/2022
- **“Challenge Cup” National College Students’ Extracurricular Academic Science and Technology Contest (Smart Grid Control)(Bronze Prize)** 07/2022